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Farmer Circle:An Online Auction Website for Selling Crops Directly to Businesses

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***Abstract*— Farmers have to face many difficulties, including the fact that they are getting paid much less than what work they do. Middle-men take advantage of the poor farmers living in villages by buying produce from them in very less price. Helping farmers by getting them a better price for their products and at the same time, helping wholesalers and other small businessmen to get fresh produce for even lower price is the main aim of our project. We have created a comprehensive website which allows a farmer and a wholesaler who buys the produce on the website. This website aims to increase the farmers' financial power by providing a bidding integrated e-commerce platform to increase their income. The farmer can start a 'bidding process' of the produce and wholesalers can bid on the produce for different prices. The farmer can then sell the produce to the highest bidder, thus getting most profit. Similarly, the bidder can choose what all goods he wants to buy, and get a fresh produce at a much less price.**

Keywords*— *Agriculture, Farmers, Auction system, Buying and selling, Bidding

I. INTRODUCTION

THIS paper talks about creating an auction system aimed at farmers, for selling and buying procure. Farmers find it difficult to sell their produce when in a village. After the harvesting period, a middle-man comes with a truck to buy the procure, and takes it away to the market to sell it. The time when the middle-man comes to the village is erratic, and the farmer has to wait for the arrival of the middle-man. Then, the farmer practically only has one choice - to sell the produce to the middle-man on the terms of the middle-man itself. The middleman usually takes advantage of this by buying if from the farmers at extremely low prices and selling at high prices in the market. Selling at high price means that the wholesalers also get the produce at unnecessary higher prices. Also, if the middle-man comes late to the village, the produce becomes stale and this deteriorates the quality of the produce, thus fetching less money for the farmer. Our auction system is aimed at minimizing all these problems, while increasing the income of the farmers.

One of the papers suggested FarmBid, an e commerce platform to provide fresh products from farmers to consumers. It focuses on contract farming, which has benefits such as better loan and credit, but can lead to exploitation of farmers from large agribusiness firms. It provides transparent market information from farmers to end consumers, simplifies the supply chain, and reduces prices. Paper 2 is on Mobile application for the purpose of marketing, product distribution and location-based logistics for elderly farmers. This research has developed a one-stop service supply chain mobile application for the purpose of marketing, product distribution and location-based logistics for elderly farmers and consumers. Paper 3 is on why agricultural policies fail and two cases of policy failures in Albania. Paper 4 - Small Farmers in India: Challenges and Opportunities. This paper examines the roles and challenges of small holding agriculture in India, covering trends in agricultural growth, cultivation patterns, participation, productivity performance, linking small holders with markets, role of small holders in enhancing food security and employment generation, differential policies and institutional support for small holders, challenges and future options for small holding agriculture, and lessons from India on small holding agriculture for other countries. Paper 5 - Market Performance of Red Chili Through Spot Auction System in Yogyakarta Province. Red chili marketing needs a spot auction system at farmer level to decrease long channel distribution and increase selling price. This analysis shows that marketing margin at spot auction market is more efficient, and that big dominances of marketing institute in red chili marketing chain influence bargaining position of red chili farmer. Paper 6 - Bidder and Seller Strategies in Online Auctions: A Review

This paper summarizes and compares existing research on the strategies for bidders and sellers in online auctions to conclude that sniping could be an optimal strategy for bidders to employ and that sellers must take a more strategic approach in selecting discretionary attributes in their auction listings to maximize profits.

We identified that these papers had various shortcomings. They were either not focused on India, but on different markets and the results were focused there. One paper was

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focused just on red chilies, but not for any general produce. By collaborating with different buyers, there are ways to exploit farmers by reducing their selling price in the ways suggested by the papers. Our paper tries to minimize these problems by providing a more comprehensive solution

II. Methodology/Experimental

A. Synthesis/Algorithm/Design/Method

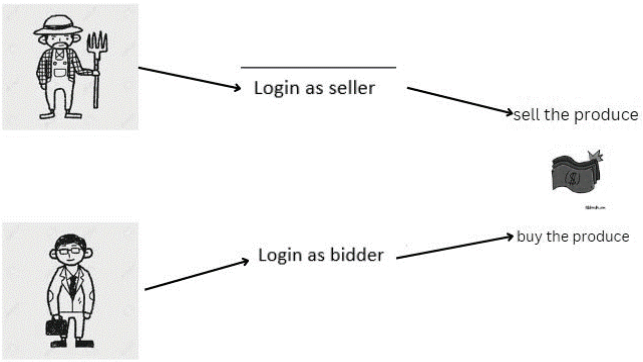


Fig. 1. High-level system flow

The design of the website is in such a way that it would be easy for anyone to use. There would be a login page, separate for both farmer and bidder. The farmer can start a bidding process for the procure. The farmer would be able to enter all the details of the procure, including how much is the weight, the quality and his location. He would also set the time for which the bidding would remain open. The wholesalers can log in as bidders. They would be able to bid on the procure for a price. At the end of the auction, the farmer would sell the procure to the highest bidder. As the farmer can sell the procure to the government at the base price, the bidders would be able to bid on the procure for a price greater than the base price.

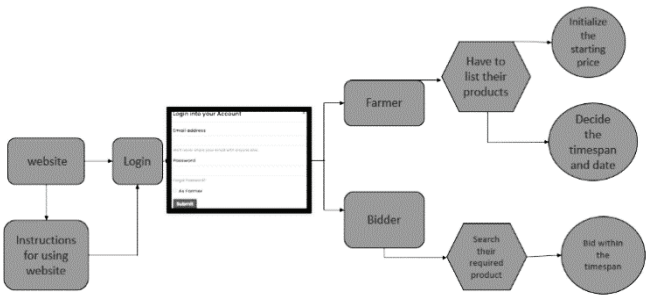


Fig. 2. Detailed user flow

The website is designed in such a way that both the farmer and the wholesalers can create a new account with their information, which includes their phone number, address and other details. After creating an account, they can log in using their credentials. Only the user registered as a farmer can an auction. While starting an auction, the farmer has to specify various details of the product and the length of the auction itself. The bidder can search through various auctions, and place a bid on an auction which he thinks is good and profitable. When the auction ends, the farmer and the highest auctioneer can see each other's' details. Then they can co-ordinate between themselves about the transportation and other logistics involved.

B. Actual implementation

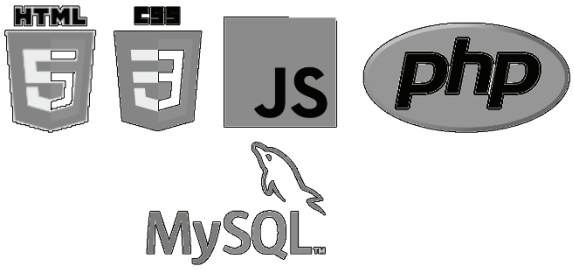


Fig. 3. Technologies used

Making of website is divided into these major tasks – HTML, CSS and JavaScript-PHP.

HTML - Layout and structure of our website is done through this. Retrieves online information via hypertext links at click of button. This creates the base skeleton of the website, on which different components, stylizing and features are added upon.

CSS - Designing, animations, transitions, etc. are done using CSS. To alter the font, color, size and spacing of contents and other decorative features. This plays a very important role in our website as we want the website to be as user-friendly as possible so that the farmers living in villages, too, should be able to navigate and use the website.

JavaScript - PHP- PHP is responsible for handling things on server and database side, and acting as a joining link between MySQL database at the backend and the front-end of the website. All the SQL queries are called through PHP. JavaScript is responsible for handling things on client side of browser.

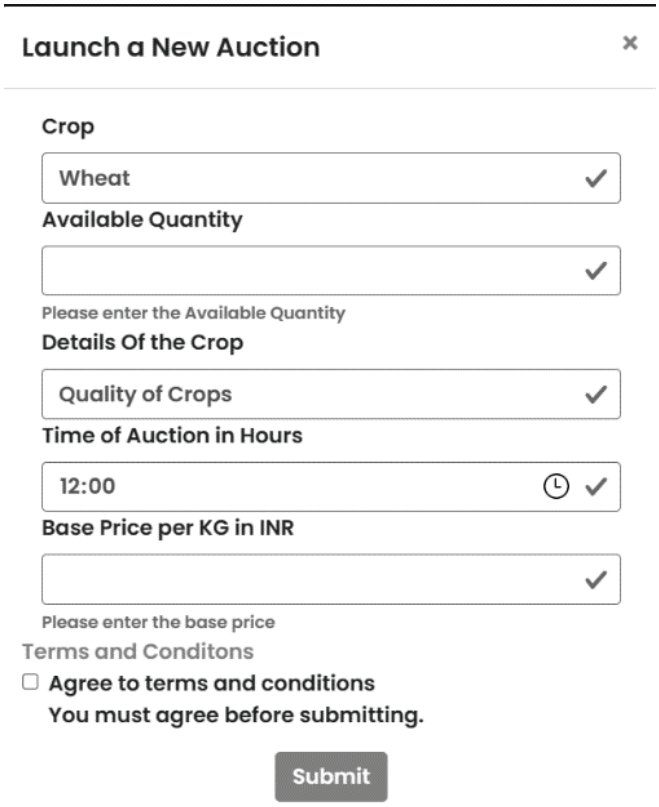


Fig. 4. Launching a new auction

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Additionally, we used the XAMPP server for the development purposes, through which we can set up a localhost connection and run the website with all of its components.

This is how the page showing auction creation looks like. The farmer can adjust a variety of parameters, including the crop type, the available quantity, the time for which the auction would remain open and the base price the crop would fetch off-auction.

Our website also focuses on the fact that farmers from villages would be using it. Thus, focusing on simple and easy-to-understand User Interface was an important part of it. Support for local languages like Marathi has also been added to the website, making it even accessible and easy-to-approach for users not accustomed with English.

III. Results and Discussions

The end product helps to improve the accessibility of the market as well as make awareness in farmers about the accurate and latest market information by providing a transparent market information platform to the farmers and end consumers. For farmers who wish to sell their produce directly to consumers, a farmer auction website can be a useful platform. It may result in better market accessibility, greater produce pricing, and increased selling efficiency. To raise demand for their produce and increase their profitability, farmers can upload information about their produce, including its quality, quantity, and price, to the website. Farmers can benefit significantly from using a farmer auction website, including better market accessibility, increased produce pricing, and higher productivity. When buyers and farmers may check out each other's profiles and reviews before making a deal, it can also increase transparency and credibility in the selling process. Possible difficulties include practically putting a farmer auction website into action.

IV. Future Scope

The future scope of the project includes adding various Indian local languages to make the website even more friendly for the farmers. Adding a quality checking mechanism would help too. Also, adding various sorting mechanisms for the bidders like sorting by price or by region would make the process smoother and easier. On the technical side, moving the project to NodeJS framework and using MongoDB for database instead of MySQL is included in the future scope, keeping in mind the scalability of the project. Also, affiliating with the company to sell their pesticides and herbicides to farmers. Also, we will facilitate with the delivery and logistics parts, helping both the farmers and wholesalers even more.

V. Conclusion

Farmers in India have to face a lot of problems. Getting less price for their produce, this greatly weakens the system as a whole, with middlemen squeezing the majority of the profit. Using technology to solve this problem can help it reduce to a large extent, though we cannot eliminate it completely as, in the end, everyone stills tries to maximize the profit of themselves. Our solution is a website. This method approaches the problem by contacting the farmers and the wholesalers directly, and creating an auction system between them of the produced goods and procure. We have successfully created a live auction system designed specifically for Indian farmers. Also, successfully conducted a live auction for a small scale while testing it. We found the results very satisfying, and plan on scaling up.

Acknowledgment

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